PW26_BJD_21 ESA PROJECT STUDY NOTE

The problem

Our project studies simulated agents that search for food and return it to a nest. Agents communicate indirectly through shared food and home pheromone fields. This is stigmergy. A compromised agent can add false food pheromone that looks like an ordinary trail. The research question is whether local signal history and own task progress can limit disruption without reducing normal retrieval.

The gap and scope

False-pheromone attacks and cautionary mitigation already exist. Learned stigmergy also exists. Our candidate gap is a controlled evaluation of parameter-shared PPO and local defense with matched trajectories, leakage boundaries and unseen conditions. We compare eight core papers here; the broader thirty-plus survey is incomplete. Our evidence is a grid simulation, not a physical robot deployment.

How the system works

Candidate J uses a 12 by 12 grid, four agents, a 2 by 2 nest and two adjacent distant food cells with fifty units each. Each agent sees only a 3 by 3 patch. The patch has six channels per cell, plus four own-state flags: 58 inputs in total. One shared PPO chooses stay, north, east, south or west. The simulator handles movement, pickup/delivery, ordinary deposits, permitted injection, clipping and evaporation in that order.

Learning and clean evidence

PPO learns from simulator interactions and shared delivery rewards, not a labelled action dataset. Full PPO and a separate no-pheromone PPO train from scratch with matched settings; the latter masks eighteen entries throughout training and evaluation. Full means are 100, 80 and 100 for seeds 7, 17 and 27. No-pheromone means are 68.75, 63.50 and 80.50. Each averages four maps. Deterministic full means are mixed: 100, 0 and 60.

DATA EVIDENCE AND NEXT STEPS

Attack results and honest explanation

One selected attacker out of four is 25%. Loss equals honest delivery with injection disabled minus honest delivery when attacked. Negative loss means higher attacked delivery in that experiment; it does not prove general harmlessness. Old mean losses were -2.17, -0.42 and -2.25. Stronger sustained injection, three units per step with cap 3,300, gives +10.53 mean loss: 70.33 to 59.81, about 15%. Only 18 of 36 pairs lose food, so the consistency gate still fails. The separate decoy route also failed to add mean injection damage.

Dataset strategy

The selected strategy is simulator-generated matched clean, injection-disabled and attacked episodes. The recorded historical corpus has 36 groups and 108 episodes. Four agent rows for each of 103,004 world transitions give 412,016 CSV rows. The CSV has 58 observations, twelve local-history features, ten provenance columns and nine outcomes. PPO still uses only 58 inputs. New diagnostic episodes are separate and do not automatically change the old corpus counts.

What keeps the evaluation fair

Use frozen policies, matched maps/seeds and complete scenario families. Keep attacker identity, injection mass and global information in audit records, never detector inputs. Own local history must use information available before the action. Split maps and related episode families before making temporal windows; randomly mixing rows would leak correlated experiences into testing.

What is complete and what comes next

The simulator, PPO, clean comparisons, attack harness, corpus builder and CSV exporter work. The recorded 6 November suite passed 172 tests. A trained detector, final exposure labels, grouped train/validation/test partitions, windows, learned-defense runs and held-out results are not complete. Next: independently confirm the attack, define residual-exposure labels, freeze grouped partitions, then train and evaluate a local detector and mitigation against the required baselines. Do not claim the simulator knowing the attacker is already a finished label definition.